

Harkirat Singh

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Professional Experience

Serigor Inc.

Columbia, MD

Data Analyst

June 2025 - December 2025

- Processed and validated Maryland tax datasets, reducing data inconsistencies by 15% and improving reporting accuracy
- Analyzed revenue trends and taxpayer behavior using Python, SQL, and Excel, supporting data-driven decisions that improved reporting efficiency by 20%
- Developed Power BI dashboards to visualize tax and operational data, improving reporting efficiency by 20%
- Gathered business requirements and translated them into SQL queries, reducing report turnaround time by 25%

Mokxa Technologies, LLC

Columbia, MD

Software Developer

January 2024 - February 2024

- Developed Python backend components and REST APIs to support enterprise banking applications
- Assisted in data validation and system testing, improving data accuracy by 15%
- Collaborated with senior developers in an Agile Scrum environment, reducing issue resolution time by 20%
- Worked with HTML and CSS to enhance user interface functionality and user experience

Infotrend Inc.

College Park, MD

Information Security Analyst

June 2023 - August 2023

- Analyzed network traffic and TCP packet data using Wireshark to identify communication patterns and support security investigations
- Presented findings on security testing methodologies and cybersecurity tools to technical and non-technical stakeholders
- Assisted in incident response and security monitoring activities, contributing to improved threat detection and response processes
- Automated routine security analysis tasks using Python scripts, improving team efficiency by 15%

Projects

AI Financial Analyst Copilot

<https://ai-stock-analyst-pal.streamlit.app/>

- Developed an AI-powered financial analysis copilot using Streamlit, OpenAI, and Yahoo Finance APIs to generate real-time investment insights and natural language market summaries
- Integrated automated data ingestion, KPI calculations, and multi-ticker comparison capabilities, enabling users to analyze and benchmark up to five stocks simultaneously
- Designed an end-to-end analytics workflow with modular architecture and secure API key management, preparing the application for future AWS deployment and scalable enterprise use

AWS Retail Sales Data Pipeline Project

<https://github.com/presiSingh99/aws-retail-sales-pipeline>

- Built an end-to-end AWS data pipeline using Amazon S3, Python (Polars), boto3, and Amazon Athena to ingest, transform, and analyze retail sales data
- Developed ETL workflows to standardize datasets, remove duplicates and null values, enforce data quality checks, and generate analytics-ready outputs
- Designed raw and processed data lake layers in Amazon S3 and enabled serverless SQL reporting through Athena to derive business insights from sales transactions
- Troubleshooted and resolved cloud integration challenges involving IAM permissions, dependency conflicts, and Athena schema mismatches to ensure reliable pipeline execution

Automated Sales Data Pipeline

github.com/presiSingh99/project1_p1_de

- Built an automated ETL pipeline using Python, Polars, and DuckDB, reducing processing time by 50%
- Implemented data validation, error handling, and debugging workflows to ensure reliable processing and reporting
- Automated multi-stage data workflows, reducing manual processing effort by 30% and improving data quality
- Designed the solution for cloud migration by structuring the pipeline for future AWS S3 integration

Hospital Readmission Prediction System

github.com/presiSingh99/ml_readmission_project1

- Developed a predictive modeling pipeline using Python, Pandas, NumPy, and Scikit-learn on 101K+ patient encounters
- Applied ensemble and supervised learning models to discover data patterns and maximize predictive performance
- Implemented PCA for dimensionality reduction, conducted hyperparameter tuning, and evaluated model performance using accuracy, precision, recall, and F1-score.

Technical & Product Competencies

Data & Analytics: Python, Java, SQL, ETL Pipelines, Data Cleaning, Data Transformation, EDA, Pandas, NumPy, Polars, DuckDB, Power BI

AI Tools: ETL Pipeline Automation, KPI Development, Generative AI Implementation, Sentiment Analysis (LLM Agents), Workflow Orchestration

Cloud & Tools: AWS (S3), Databricks, Azure AI Foundry, GitHub, RestAPI

Education

University of Maryland, Baltimore County

Baltimore, MD

Bachelor of Science, Information Systems

Graduated May 2026

